

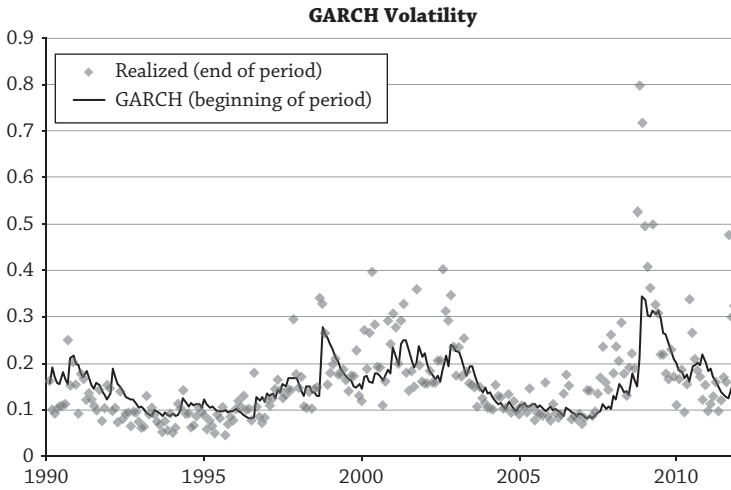


Figure 8.8

(Lehman default)—lead to higher future volatility. Once these hit, volatility remains high for a certain period of time before mean-reverting. This allows the GARCH model to capture periods of turbulence and quiet. We see this in Figure 8.8, which plots GARCH forecasted volatility with realized volatility (computed using realized daily returns over the month), both at the monthly frequency, from January 1990 to December 2012. The correlation of GARCH volatility at the beginning of the month with realized volatility over the next month is 63%. Compare this correlation with the barely 5% correlations that we were expecting to predict future equity risk premiums! Volatility is quite predictable in equity markets.

If volatility is so predictable, then volatility trading should lead to terrific investment gains. It does. Despite my pessimism on predicting expected returns of the previous section, I am far more enthusiastic on strategies predicting volatilities.

Panel A of Figure 8.9 contrasts cumulated returns from January 1986 to December 2011 of a static 60% equities–40% T-bill strategy with a volatility timing strategy based on VIX, which is also overlaid on the graph. The static strategy is the mean-variance portfolio weight given in equation (2.10) in chapter 2, which for convenience is repeated here:

$$w = \frac{1}{\gamma} \frac{\mu - r_f}{\sigma^2}, \quad (8.9)$$

where the risk aversion, γ , is calibrated to produce a 60%/40% portfolio over the whole sample. The volatility timing strategy replaces market volatility, σ , in the denominator of equation (8.9) with the VIX, so that it becomes time varying.